

Project Plan Documentation

Project Title

FarmVense Nexus – Precision Agriculture Management System using Full Stack Java

1. Introduction

FarmVense Nexus is a Full Stack Java-based web application designed to help farmers manage agricultural activities digitally. The system provides a centralized platform for managing farms, crops, irrigation, fertilizer usage, expenses, harvest details, weather updates, notifications, and reports.

The main aim of this project is to reduce manual record-keeping and help farmers make better decisions using organized farm data.

2. Purpose of the Project

The purpose of FarmVense Nexus is to provide a smart and simple agriculture management system for farmers. It helps farmers store farm information, monitor crop activities, schedule irrigation, track expenses, and calculate profit or loss.

This system supports farmers in improving productivity, reducing resource wastage, and managing farming activities more efficiently.

3. Objectives

The main objectives of the project are:

- To develop a Full Stack Java web application for agriculture management.
- To provide secure login for farmers and administrators.
- To manage farm, crop, irrigation, fertilizer, expense, and harvest details.
- To reduce manual farm record maintenance.
- To help farmers track expenses and calculate profit or loss.
- To provide weather information and activity reminders.
- To generate useful reports for better decision-making.
- To design a responsive and user-friendly dashboard.

4. Scope of the Project

FarmVense Nexus includes two main users: **Farmer** and **Admin**.

Farmer

The farmer can:

- Register and log in.
- Add and manage farm details.

- Add and update crop information.
- Schedule irrigation activities.
- Record fertilizer and pesticide usage.
- Track farm expenses.
- Add harvest details.
- View weather updates.
- Receive notifications.
- Generate reports.

Admin

The admin can:

- Manage farmer accounts.
- View farm and crop statistics.
- Manage system data.
- Monitor overall application activities.
- Generate system-level reports.

5. Technologies Used

Frontend

- HTML
- CSS
- JavaScript
- Bootstrap

Backend

- Java
- Spring Boot
- Spring MVC
- REST APIs
- Spring Security

Database

- MySQL

ORM and Tools

- Hibernate
- Spring Data JPA
- Maven
- Postman
- GitHub
- Eclipse / IntelliJ IDEA

6. Project Modules

6.1 User Management Module

This module manages registration, login, authentication, and role-based access.

Main features:

- Farmer registration
- Admin and farmer login
- Password security
- Profile management
- Logout

6.2 Farm Management Module

This module allows farmers to manage farm details.

Main features:

- Add farm name
- Add farm location
- Add land area
- Add soil type
- Add water source
- Update or delete farm details

6.3 Crop Management Module

This module manages crop information from sowing to harvesting.

Main features:

- Add crop name and crop type
- Add sowing date
- Add expected harvest date
- Track crop growth stage
- Update crop status

6.4 Irrigation Management Module

This module helps farmers plan and manage water usage.

Main features:

- Schedule irrigation
- Record irrigation date
- Track water quantity
- Select irrigation method
- View irrigation history

6.5 Fertilizer and Pesticide Management Module

This module stores fertilizer and pesticide usage details.

Main features:

- Add fertilizer and pesticide details
- Track quantity used
- Record usage date
- Track cost
- Maintain usage history

6.6 Expense Management Module

This module helps farmers maintain farming expense records.

Main features:

- Add seed cost
- Add fertilizer cost
- Add pesticide cost
- Add labor cost
- Add machinery cost
- Calculate total expenses

6.7 Harvest Management Module

This module stores harvest-related information.

Main features:

- Add harvest date
- Add crop quantity
- Add selling price
- Calculate total income
- Compare income and expenses

6.8 Weather and Notification Module

This module provides weather information and important reminders.

Main features:

- View temperature and humidity
- View rainfall information
- Receive weather alerts
- Receive irrigation reminders
- Receive fertilizer and harvest reminders

6.9 Reports and Analytics Module

This module generates reports based on farm data.

Main features:

- Crop report
- Irrigation report
- Expense report
- Harvest report
- Profit or loss report

7. System Architecture

FarmVense Nexus follows a three-layer architecture:

Presentation Layer

This layer contains the frontend pages where users interact with the application.

Business Logic Layer

This layer contains backend logic developed using Java and Spring Boot.

Database Layer

This layer stores all user, farm, crop, irrigation, expense, and harvest data using MySQL.

Architecture Flow

Frontend → REST API → Spring Boot Backend → MySQL Database

8. Database Tables

The main database tables are:

- Users
- Roles
- Farms
- Crops
- Irrigation
- Fertilizers
- Pesticides
- Expenses
- Harvest
- Weather
- Notifications
- Reports

9. Functional Requirements

The functional requirements of the system are:

- The system should allow farmer registration and login.
- The farmer should be able to manage farm and crop details.
- The farmer should be able to schedule irrigation.
- The farmer should be able to record fertilizer and pesticide usage.
- The farmer should be able to add expenses and harvest details.
- The system should calculate income, expenses, profit, and loss.
- The system should provide weather updates and reminders.
- The system should generate reports.
- The admin should be able to manage users and system data.

10. Non-Functional Requirements

The non-functional requirements are:

- The system should be secure.
- The system should be easy to use.
- The system should be responsive.
- The system should provide fast performance.
- The system should store data safely.
- The system should be scalable.
- The user interface should be simple and clean.

11. Testing Plan

Testing will be performed to ensure that the system works correctly.

Unit Testing

Individual modules such as login, farm, crop, irrigation, and expense modules will be tested separately.

Integration Testing

Related modules will be tested together to check proper data flow.

Example:

- User module with farm module
- Crop module with irrigation module
- Expense module with harvest module

System Testing

The complete application will be tested from registration to report generation.

User Acceptance Testing

The system will be tested from the farmer's point of view to check whether it is simple and useful.

12. Project Timeline

Phase	Task	Duration
Phase 1	Requirement Analysis	1 week
Phase 2	System and Database Design	1 week
Phase 3	Frontend Development	2 weeks
Phase 4	Backend Development	3 weeks
Phase 5	Database Integration	1 week
Phase 6	Testing and Bug Fixing	1 week
Phase 7	Documentation and Submission	1 week

Total estimated duration: **10 weeks**

13. Expected Output

The expected output of the project is a fully functional agriculture management web application.

The system will provide:

- Farmer dashboard
- Farm management
- Crop management
- Irrigation scheduling
- Fertilizer and pesticide tracking
- Expense tracking
- Harvest management
- Weather updates
- Notifications
- Reports and analytics

14. Future Enhancements

The project can be enhanced in the future with:

- IoT-based soil moisture monitoring
- AI-based crop disease detection
- Machine learning-based yield prediction
- Drone-based crop monitoring
- SMS alerts for farmers
- Farmer marketplace
- Government scheme updates

15. Conclusion

FarmVense Nexus is a useful Full Stack Java project for digital farm management. It helps farmers manage farms, crops, irrigation, fertilizers, expenses, harvest records, and reports from a single platform.

The system reduces manual work, improves record maintenance, supports better decision-making, and promotes efficient farming practices.